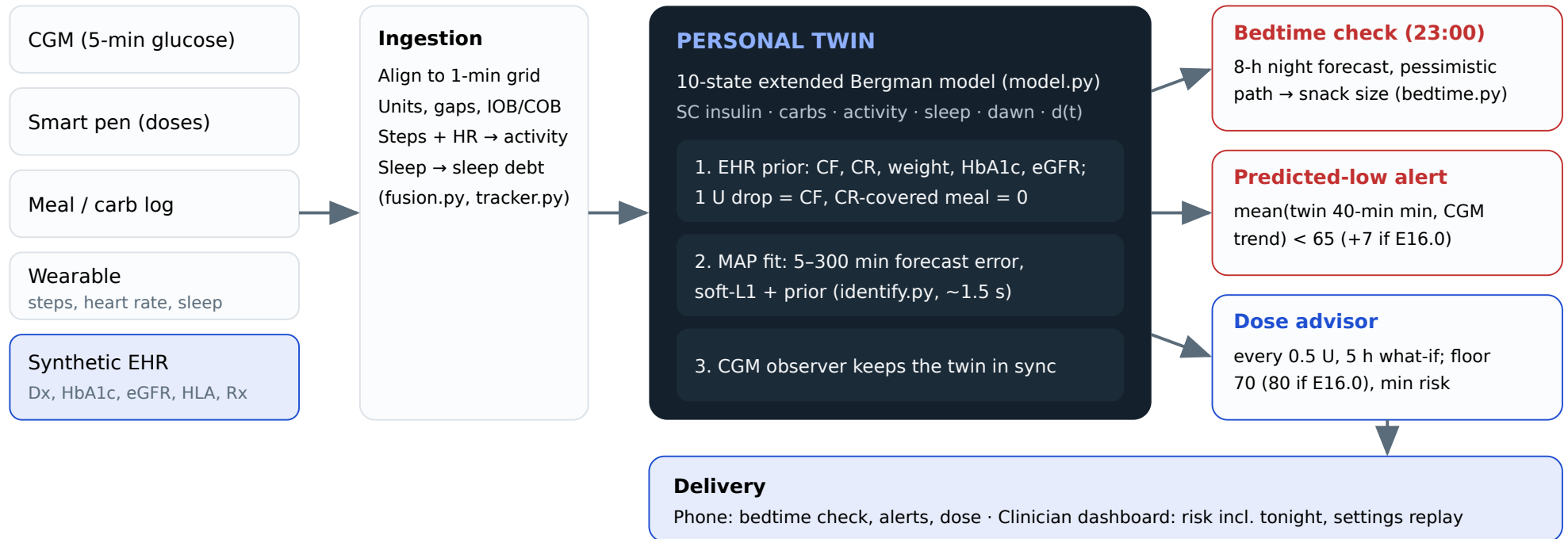


DoseTwin: system architecture

Personal digital twin for insulin-treated diabetes (validated on type 1, on insulin pens). Solid boxes run in deployment; the dashed band is the in-silico evaluation harness.

DATA SOURCES



IN-SILICO EVALUATION HARNESS (not part of deployment)

Hidden truth: UVA/Padova T1D model + sleep, kidney-clearance and hypo-awareness effects, 30 virtual patients (parity vs simglucose <0.5 mg/dL)

Life simulator: Indian meal timing, carb-count bias + 30% error, missed/late boluses, mis-set settings, exercise, short nights, CGM noise (cohort.py)

Protocol: dev cohort (seed 7) for tuning + LightGBM baseline; held-out seeds 2026-2028; days 1-7 run-in, days 8-14 paired arms A-G, refitted ablations

Outputs: results/study.json, CSVs, figures — bootstrap CIs, Wilcoxon tests (run_study.py, ~25 min CPU)